SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1519

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 20**
Question Count: 4

Code of Conduct

I S.MAGALAKSHMI (192524144) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S Magalakshmi

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Cloud Computing Fundamentals

Definition

Cloud Computing is the delivery of computing resources (Servers, Storage, applications and Services) over the internet on a pay-as-you-go basis

Cloud Computing



Characteristics

- On-demand Self-Service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured Service

Benefits

- Cost efficiency
- Scalability & Flexibility
- High availability
- Access from anywhere
- Reduced maintenance

Limitations

- Security and privacy Concerns
- Dependence on internet
- Downtime risk
- Limited Control
- Data transfer costs

Evolution Of Cloud Computing

Evolution Of Cloud Computing

Hardware Evolution

- Mainframes
- Personal Computers
- Servers & Clusters
- Virtualized hardware

More powerful and efficient infrastructure

Internet & Software Evolution

- Web (HTTP/HTML)
- Web 2.0
- APIs & Web Services
- Mobile & IoT

Connected world & richer applications

Distributed System

- Distributed Computing
- Grid Computing
- Fault tolerance
- Load balancing

Multiple Computers Work as One

Service Delivery

IaaS (infrastructure)
PaaS (Platform)
SaaS (applications)
xaaS (anything as a Service)

Services delivered via the internet

Result

Modern, Scalable, flexible and on-demand cloud Computing

Service Models



Different levels of abstraction for delivering cloud Service

IaaS
(Infrastructure as a Service)

Provides virtualized computing resources (vms, Storage, networks)

Provider Responsibility
- Physical hardware
- Virtualization
- Infrastructure
- Security

User Responsibility
- Applications
- Data

PaaS
(Platform as a Service)

Provides development platform (runtime, middleware, databases, tools)

Provider Responsibility
- Infrastructure
- Platform management
- Runtime
- Security

User Responsibility
- Applications
- Data
- Configuration

SaaS
(Software as a Service)

Provides ready-to-use applications e.g., emails, CRM, office tools

Provider Responsibility
- Infrastructure
- Platform
- Application
- Security

User Responsibility
- Data
- Application Settings
- User management

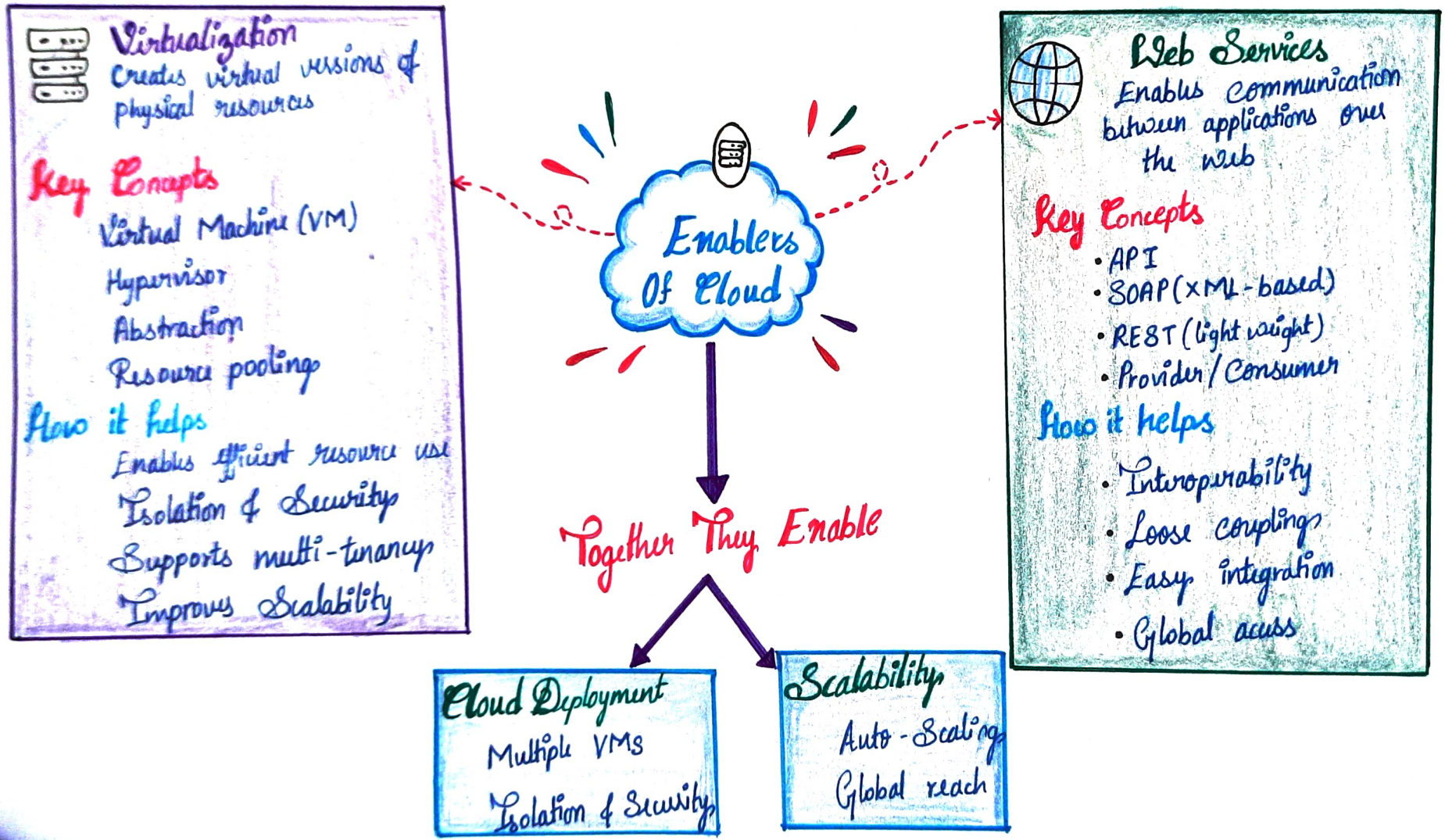
XaaS
(Anything as a Service)

Includes e.g., DBaaS, FaaS, CaaS, NaaS, etc.

Provider Responsibility
- Infrastructure
- Platform
- Security

User Responsibility
- Data
- Configuration
- Usage/Access

Virtualization and Web Services as Enablers Of Cloud



Cloud Computing Fundamentals

Cloud computing is a technology that provides computing resources such as Servers, Storage, applications and Services through the Internet. Instead of maintaining all the hardware and Software locally, users can access the required resources whenever needed and generally pay according to their usage. Cloud computing has several important characteristics, including on-demand Self-Service, broad network acuss, resource pooling, rapid elasticity and measured Service. On-demand acuss allows users to obtain resources whenever required, while resource pooling allows computing resources to be Shared among multiple users. Rapid elasticity allows resources to increase or decrease according to demand, and measured Service helps monitor and measure resource usage. Cloud computing provides many benefits such as cost efficiency, Scalability, flexibility, high availability, anywhere acuss and reduced maintance. However, it also has limitations, including Security and privacy concerns, dependence on internet connectivity, downtime risk, limited control over infrastructure and data-transfer costs.

Evolution Of Cloud Computing

The evolution of cloud computing is the result of continuous development in hardware, Internet and Software technologies, distributed Systems and Service delivery. In the early stages, computing was mainly based on mainframes, followed by the development of personal computers. Later, powerful Servers and clusters were introduced, and virtualization made it possible to create multiple virtual resources on physical hardware. At the same time, Internet and Software technologies developed from basic Web technologies using HTTP and HTML to Web 2.0, APIs, web Services, mobile technologies and IoT. Distributed Systems also developed, allowing multiple computers to work together through distributed computing and grid computing, while concepts Such as fault tolerance and load balancing improved reliability and performance. These developments gradually made it possible to deliver computing resources as Services through the Internet. This evolution finally resulted in modern cloud computing, which provides Scalable, flexible, on-demand and Shared computing Services.

Service Models

Cloud computing offers different Service models that provide different levels of computing resources and resources and responsibilities to users. The major models are IaaS, PaaS, SaaS and xaaS. Infrastructure as a Service (IaaS) provides basic computing infrastructure such as virtual machines, storage and networks, where the provider manages the physical infrastructure and virtualization while the user manages the operating System, applications and data. Platform as a Service (PaaS) provides a development platform including runtime, middleware, databases and development tools, allowing users to focus mainly on applications and data. Software as a Service (SaaS) provides ready-to-use applications such as email, CRM and office applications, with most of the infrastructure, platform and application management handled by the provider. xaaS (Anything as a Service) represents various cloud-based Services such as DBaaS, FaaS, CaaS and NaaS. Across these models, responsibilities are divided between the cloud provider and the consumer, forming a Shared responsibility approach.

Virtualization & Web Services as Enablers Of Cloud

Virtualization and Web Services are important technologies that enable cloud computing. Virtualization creates virtual versions of physical computing resources and allows a single physical machine to run multiple virtual machines (VMs). A hypervisor manages these virtual machines and provides abstraction between physical hardware and virtual resources. Virtualization support resources pooling, efficient resource utilization, isolation, security and scalability, making cloud deployment easier and more flexible. Web Services, on the other hand, allow different applications and systems to communicate with each other through the Internet. They use technologies such as APIs, SOAP and REST. SOAP is an XML-based web service approach, while REST provides a lightweight method for communication between applications. Web Services involve a provider and consumer, where the provider offers a service and the consumer uses it. They support interperability, loose coupling, easy integration and global access. Therefore, virtualization and Web Services work together to support cloud deployment, resource sharing and scalability, resulting in efficient, flexible and scalable cloud solutions.